

| Question number | Answer                                                                                                                                                                                                                                                                | Additional guidance                                                                                                                                                                                        | Mark |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 7(a)            | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>because MRSA has developed {antibiotic / methicillin} resistance (1)</li> <li>so we have had to find new drugs that this bacteria is not resistant to (1)</li> </ul> | <p><b>NB ACCEPT</b> pathogens / bacteria throughout</p> <p><b>DO NOT ACCEPT</b> immune for resistance throughout</p> <p><b>ACCEPT</b> spreading / increasing</p> <p><b>ACCEPT</b> that will destroy it</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                        | Additional guidance                                                                                                                                  | Mark |
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| 7(b)            | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>because skin flora {prevent the growth of / destroy / kill} {other bacteria / pathogens} (1)</li> <li>by competing for {space / nutrients} / producing toxins (1)</li> </ul> | <p><b>ACCEPT</b> provide a barrier in protecting the body from infection</p> <p><b>ACCEPT</b> produce lactic acid<br/><b>DO NOT ACCEPT</b> sebum</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                                                                                            | Additional guidance                                                                                                                     | Mark |
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| 7(c)            | <p>An answer that includes two of the following points:</p> <ul style="list-style-type: none"> <li>(if the bacteria are weaker) they will {not reproduce as fast / infect fewer cells} (1)</li> <li>therefore there will be fewer toxins produced (1)</li> <li>therefore an immune response will be more effective (1)</li> </ul> | <p><b>ACCEPT</b> descriptions e.g. macrophages will be able to engulf more, more antibodies produced, more T killer cells activated</p> | (2)  |

| Question number | Answer                                                                                                                                                                                                                                                        | Additional guidance                                                                      | Mark       |
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| 7(d)(i)         | <p>An explanation that includes the following points:</p> <ul style="list-style-type: none"> <li>• so that the size of the infection with the solvent could be measured (1)</li> <li>• so that the effect of the drug itself can be determined (1)</li> </ul> | <p><b>ACCEPT</b> to show {the effect of Castaneroxy / that Castaneroxy is effective}</p> | <b>(2)</b> |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Additional guidance                                                                                                                                                                                                                                                                                                                            | Mark       |
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| 7(d)(ii)        | <p>An answer that includes any <b>two</b> of the following points:</p> <ul style="list-style-type: none"> <li>comment about effect of Castaneroxy on size during time period 0 and 6 days (1)</li> <li>comment made about Castaneroxy (compared to solvent) on size during time period {0 / 6} and 14 days (1)</li> <li>comment comparing the two concentrations of Castaneroxy (1)</li> </ul> <p><b>AND</b></p> <ul style="list-style-type: none"> <li>credit calculation to quantify reduction in size (1)</li> </ul> | <p>e.g. Castaneroxy does not prevent increase in size, the size does not increase as much as the solvent</p> <p>e.g. Castaneroxy reduces mean size</p> <p>e.g. Castaneroxy at high concentration decreases the size more than at lower concentrations, high concentrations clear up the infection completely but low concentrations do not</p> | <b>(3)</b> |

| Question number | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                        | Additional guidance                                                              | Marks |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|-------|
| (iii)           | <ul style="list-style-type: none"> <li>mass of Castaneroxy A needed in 1 dm<sup>3</sup> calculated (1)</li> <li>mass of leaves needed for 50 000 µg (1)</li> <li>converted into kg and rounded up to 3 (kg) (1)</li> </ul> <p style="text-align: center;"><b>OR</b></p> <ul style="list-style-type: none"> <li>50 µg converted into kg (1)</li> <li>mass needed in 1 litre (1)</li> <li>converted into kg and rounded up to 3 (kg)</li> </ul> | 50 000<br><br>2 631 578 947<br><br><br><br><br><br><br>0.00000005<br><br>0.00005 | (3)   |